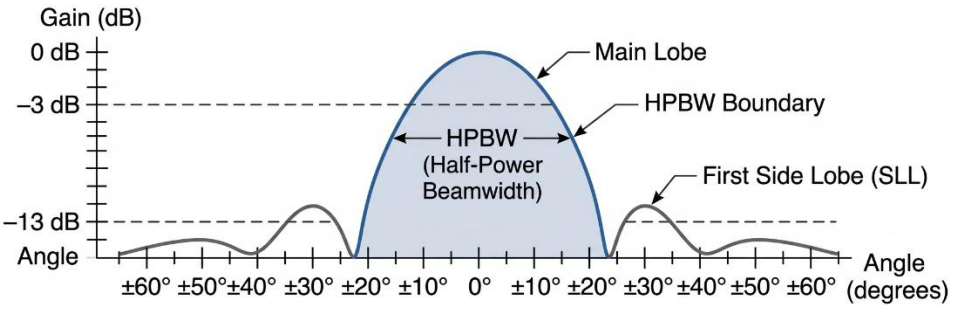
Electromagnetic Communication

1. Consider the general communication system model consisting of a Source, Transmitter, Transmission Channel, Receiver, and Destination.
    a. Identify where an Electronic Support Measures (ESM) receiver and a Stand-Off Jammer (SOJ) interact with this classic block diagram.
    b. Explain how intentional high-power noise injection in the transmission channel affects the signal-to-noise-plus-interference ratio (SNIR) at the receiver input.
2. A simple pulsed surveillance radar transmits a carrier frequency fc=10 GHz modulated by a periodic rectangular pulse train. The pulse width is τ=2 μs and the Pulse Repetition Interval (PRI) is Tp=100 μs.
    a. Calculate the radar's duty cycle D.
    b. If the transmitter produces a peak power P_{peak}=50 kW during the pulse, calculate the average transmitted power P_{avg}.
3. An Electronic Support (ES) system intercepts an unmodulated radar pulse train with a pulse duration τ=0.5 μs centered at RF carrier frequency fc=3 GHz.
    a. Determine the first null-to-null bandwidth B_{null} of the main lobe in the frequency spectrum.
    b. How does decreasing the pulse width τ to 0.1 μs (to improve radar range resolution) impact the required bandwidth of the ES receiver filter?
4. An intercept receiver captures an ASK-modulated transmitter spectrum displayed on a spectrum analyzer centered at fc=10 kHz. The discrete harmonic spectral lines are observed at intervals of 1 kHz.
    a. Determine the fundamental pulse repetition rate / bit rate (R_b) of the modulating message signal.
    b. If the main spectral peak power is measured at +92 dBμV and a continuous wave (CW) jamming baseline elevates the noise floor to +35 dBμV, calculate the dynamic Signal-to-Noise Ratio (SNR) in dB.
5. Modern combat radars use Linear Frequency Modulation (LFM/Chirp) to transmit high energy while retaining fine range resolution. A radar pulse has a pulse width T=10 μs and sweeps linearly from 9.95 GHz to 10.05 GHz.
    a. Calculate the total modulation bandwidth Δf and the frequency sweep rate K in MHz/μs.
    b. Why is LFM preferred over simple high-amplitude short pulses in low-probability-of-intercept (LPI) radar designs?
6. A threat evaluation (ESM) library classifies intercepted non-cooperative digital link waveforms (COMINT).
    Waveform A: Carrier shifts between 12 MHz and 16 MHz based on data state.
    Waveform B: Carrier remains at 14 MHz, but phase flips by 180° at bit boundaries.
    a. Identify the digital modulation type (FSK or PSK) for Waveform A and Waveform B.
    b. Explain which waveform maintains a constant envelope and how a 180° phase jump appears in temporal analysis.
7. A pulse compression radar uses Phase Modulation on Pulse (PMOP) with a 13-chip Barker code. The uncompressed pulse duration is τ=13 μs, divided into 13 sub-pulses (chips) of length t_c=1 μs each.
    a. Calculate the uncompressed range resolution ΔR_{uncomp} ($c=3\times10^8$ m/s).
    b. Calculate the compressed range resolution ΔR_{comp} after matched filtering in the receiver.
8. A uniform plane wave propagates in free space along the +z-direction. The instantaneous electric field vector is given by $\mathbf{E}(z,t)=E_0\cos(\omega t-kz)\,\mathbf{a}_y$ (y-direction).
    a. Using the Poynting vector relationship $\mathbf{P}=\mathbf{E}\times\mathbf{H}$, determine the orientation vector unit direction ($\mathbf{a_H}$) of the magnetic field H (x,y or z).
    b. Are E, H, and P mutually perpendicular?
9. A ground-based air-defense radar tracks a supersonic target at a line-of-sight range R=45 km.
    a. Calculate the round-trip signal propagation time t_{rt}.
    b. If the aircraft is moving directly toward the radar at a velocity v=2160 km/h, calculate how far (in cm) the aircraft moves during this single pulse round-trip transit time.
10. An aircraft has a physical length L=15 m. Calculate the electrical length (L/λ) of the aircraft when illuminated by
    a. A VHF early warning radar (f=150 MHz).
    b. An X-band tracking radar (f=10 GHz).
    c. Explain why stealth shaping is much more effective in reducing Radar Cross Section (RCS) at X-band than at VHF frequencies.
11. An incoming plane wave traveling toward an observer out of the page (+z-direction) has orthogonal electric field components: $E_x(t)=E_{x0}\cos(\omega t)$, $E_y(t)=E_{y0}\cos(\omega t+\Delta\phi)$

Identify the polarization state for each scenario:

a. $E_{y0}=0$, $E_{x0}>0$
b. $E_{x0}=E_{y0}$, $\Delta\phi=+90^\circ$
c. $E_{x0}=E_{y0}$, $\Delta\phi=-90^\circ$
d. $E_{y0}=2E_{x0}$, $\Delta\phi=+90^\circ$

12. A drone-mounted isotropic communications jammer emits an omnidirectional power Pt=200 W.
    a. Calculate the power flux density P_d in W/m^2 at a standoff range R=10 km.
    b. If an ESM intercept antenna has an effective capture aperture $A_e=0.05\ m^2$, calculate the total power Pr collected (received) by the antenna at 10 km range.
13. A high-power threat transmitter emits spherical wavefronts.
    a. Why can an airborne ESM receiver located at a range R=80 km treat the incident wave as a plane wavefront?
    b. Two ESM antenna sensors (interferometer) are separated by a baseline distance d=0.5 m. If the incident plane wave arrives at an incident angle θ relative to broadside, write the spatial phase difference equation $\Delta\phi$ between the two sensors as a function of wavelength λ.
14. Consider the one-way Free Space Path Loss equation (L_p). Convert this formula into logarithmic form Lp(dB) involving distance R (in km) and frequency f (in MHz). Calculate the path loss L_p(dB) for an EW intercept link operating at f=10 GHz over a distance R=30 km.
15. Perform the following conversions and arithmetic calculations:
    a. Express a radar peak transmit power Pt=2 kW in dBW and dBm.
    b. Convert a receiver sensitivity level Prx=−90 dBm to power in watts (W).
    c. An ESM receiver receives a signal power of −40 dBm. If an inline attenuator of 6 dB is added before an amplifier with a power gain of +23 dB, calculate the output signal power in dBm.
16. A radar transmits Pt=10 kW (+70 dBm) at f=1 GHz (λ=0.3 m). An aircraft at range R=30 km carries both an ESM intercept receiver and acts as a target with isotropic scattering(unity gain).
    a. Calculate the one-way path loss Lp(dB) at R=30 km.
    b. Calculate the power P_{esm} intercepted by the aircraft's ESM receiver (assuming unity isotropic antennas).
    c. Calculate the echo power P_{radar} returning to the radar receiver if the target re-radiates isotropically without gain (unity radar cross section-RCS).
17. A ground radar antenna height is h_T=50 m and an aircraft flies at altitude h_R=18 m. Calculate the maximum Line of-Sight (LOS) distance d.
18. A tactical communication/jamming system operates at f=300 MHz.
    a. Calculate the required physical length L of a quarter-wave (λ/4) vertical monopole antenna.
    b. Repeat (a) for a λ/2-dipole.
19. Refer to the normalized antenna field radiation pattern shown below:



    a. Determine approximately HPBW (Half-Power Beamwidth), FNBW (First Null Beamwidth), and SLL (Side Lobe Level).
    b. If a main lobe peak gain is G_{main}=+30 dBi and the Side Lobe Level is SLL=−20 dB relative to the main beam peak, calculate the absolute gain of the first side lobe in dBi.
20. A tracking radar transmits Right-Hand Circular Polarization (RHCP) toward a target. The reflection off a flat metallic plate target flips the helicity of the wave. What polarization must the radar's receive antenna have to capture maximum echo energy?
21. A linearly polarized ESM antenna receives a plane wave whose electric field vector is linearly polarized at an angle $\psi_i=60^\circ$ relative to the antenna's polarization axis. Calculate the Polarization Loss Factor both as a linear ratio (PLF)

and in dB (PLF_{dB}) using $PLF=|\cos\psi_i|^2$.

22. A tracking radar operates at f=10 GHz using a circular parabolic dish antenna with diameter D=1.0 m.
    a. Calculate the gain G in dBi using the aperture approximation $G=0.55(\lambda/D)^2$.
    b. Estimate the Half-Power Beamwidth (HPBW) in degrees using $HPBW\approx 70^\circ(D/\lambda)$.
    c. If the transmitter supplies Pt=500 W (+27 dBW) to the antenna, calculate the Effective Isotropic Radiated Power (EIRP) in dBW.
23. A linear phased array radar operating at λ=3 cm has element spacing d=1.5 cm ($\lambda/2$).
    a. Calculate the required progressive phase shift $\Delta\phi$ (in degrees) between adjacent elements to steer the main beam to an angle $\theta 0=30^\circ$ off broadside using $\Delta\phi=2\pi(d/\lambda)\sin\theta$.
    b. Using the antenna comparison table provided in the slides, select the best antenna type for an airborne Radar Warning Receiver (RWR) requiring wideband coverage, circular polarization, and a broad $60^\circ\times 60^\circ$ beamwidth.
24. A radar receiver front-end RF amplifier has a Noise Figure NF=3 dB and a power gain G=15 dB.
    a. Write the Noise Figure expression NF in terms of input Signal-to-Noise Ratio (Si/Ni) and output Signal-to-Noise Ratio (So/No).
    b. If an incoming radar target echo presents an input SNR (Si/Ni)=20 dB at the amplifier terminals, calculate the output SNR (So/No) in dB.
    c. What is the Noise Figure of an ideal (noiseless) amplifier?
25. A two-stage radar receiver front-end consists of:
    a. Stage 1 (Low Noise Amplifier - LNA): Gain G1=20 dB (linear 100), Noise Figure NF1=2.0 dB (linear noise factor F1=1.585).
    b. Stage 2 (Mixer/Amplifier): Gain G2=10 dB, Noise Figure NF2=7.0 dB (linear noise factor F2=5.012).
    c. Calculate the equivalent noise temperature T_{e1} of Stage 1 alone assuming reference room temperature T_0=290 K.
    d. Calculate the total system noise factor F_{sys} .
    e. Convert F_{sys} to overall system Noise Figure NF_{sys} in dB.